# PROJECT REPORT

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# PERSONAL EXPENSE TRACKER

## A Simple Web-Based Expense Management Application

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Technologies Used: HTML, CSS, JavaScript

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# 1. Introduction

The Personal Expense Tracker is a simple web application developed to help users manage their daily expenses. Users can enter expense details such as the expense name, amount, category, and date.

The application displays the recorded expenses and automatically calculates the total amount. Users can also delete expenses when necessary.

HTML is used to create the structure of the website, CSS is used to design the user interface, and JavaScript is used to implement the main functions. The application uses browser localStorage to save expense records so that they remain available after refreshing the page.

# 2. Objectives

The main objectives of this project are:

- To record daily expenses easily.
- To store expense names, amounts, categories, and dates.
- To calculate the total expenses automatically.
- To display all recorded expenses.
- To allow users to delete expenses.
- To save expense records using localStorage.
- To create a simple and user-friendly interface.

# 3. Technologies Used

## 3.1 HTML

HTML stands for HyperText Markup Language. It is used to create the structure of the website.

In this project, HTML provides the input fields, category selection, buttons, total expenses section, and expense list.

## 3.2 CSS

CSS stands for Cascading Style Sheets. It is used to improve the appearance of the website.

In this project, CSS is used to add colours, spacing, borders, rounded corners, button styles, and a responsive layout.

## 3.3 JavaScript

JavaScript is used to make the website interactive.

In this project, JavaScript is responsible for adding expenses, displaying expense details, calculating the total amount, and deleting expenses.

### 3.4 localStorage

localStorage is a browser feature that allows data to be stored locally.

In this project, localStorage is used to save expense records. The application loads the saved records when the page is opened again. Therefore, the records remain after refreshing the page in the same browser and website origin.

# 4. System Features

### 4.1 Add Expense

Users can enter an expense name, amount, category, and date. After clicking the Add Expense button, the expense is added to the expense list.

### 4.2 Expense Categories

Users can select a category for each expense. The available categories are:

- Food
- Travel
- Education
- Other

### 4.3 View Expense List

The application displays the recorded expenses with their names, amounts, categories, and dates.

### 4.4 Calculate Total Expenses

The application automatically calculates and displays the total amount of all recorded expenses.

### 4.5 Delete Expense

Users can delete an expense using the Delete button. The expense list and total amount are updated automatically.

### 4.6 Save Expense Records

The application uses localStorage to save expense records. The saved records remain available after a normal page refresh.

### 4.7 User-Friendly Design

The website uses a simple layout, clear input fields, and styled buttons to make the application easy to use.

# 5. System Design and Implementation

The project consists of three main files.

index.html: This file contains the structure of the website, including the input fields, buttons, total expenses section, and expense list.

style.css: This file controls the appearance of the website, including colours, spacing, borders, buttons, and responsive styles.

script.js: This file contains the JavaScript functions used to add, display, delete, and save expenses.

The application stores expenses in a JavaScript array. When a user adds an expense, the application creates an expense object and adds it to the array. The display function updates the expense list and calculates the total. The save function stores the array in localStorage.

# 6. How the Application Works

The application follows these steps:

1. The user opens the website in a browser.
2. The user enters the expense name and amount.
3. The user selects a category and enters the date.
4. The user clicks the Add Expense button.
5. The application checks the required information.
6. The new expense is added to the expense list.
7. The total expenses amount is calculated automatically.
8. The expense records are saved in localStorage.
9. When the page is refreshed, the saved expenses are loaded again.
10. The user can delete an expense when necessary.

# 7. Testing

The application can be tested using the following test cases.

Test Case| Expected Result

Add a valid expense| The expense appears in the list.

Add multiple expenses| All expenses are displayed.

Check the total amount| The total is calculated correctly.

Delete an expense| The selected expense is removed.

Refresh the page| Previously saved expenses remain visible.

Enter incomplete information| A validation message is displayed.

The application was tested by adding an expense and refreshing the page. The saved expense remained visible after refreshing.

# 8. Advantages

The main advantages of the Personal Expense Tracker are:

- It is simple and easy to use.
- It helps users record daily expenses.
- It calculates total expenses automatically.
- It allows users to delete unwanted records.
- It saves expense records in the browser.
- It uses basic web development technologies.
- It provides practical experience in HTML, CSS, and JavaScript.

# 9. Limitations

The current version has some limitations:

- Data is stored in the current browser rather than an online database.

- Records may be lost if the browser's site data is cleared.

- The application does not have a user login system.

- Expenses cannot be synchronised automatically across different devices.

- Monthly reports and charts are not currently available.

# 10. Future Improvements

The application can be improved in the future by adding:

- An edit expense feature.

- A category filter.

- Monthly expense summaries.

- Charts to visualise spending.

- An option to export expense records.

- A database and user login system for online storage.

These improvements could make the application more useful and suitable for managing larger amounts of expense data.

# 11. Conclusion

The Personal Expense Tracker is a simple web application developed using HTML, CSS, and JavaScript. It allows users to add, view, and delete expenses while calculating the total amount automatically.

The use of localStorage allows the application to keep expense records after refreshing the page. The project helped develop practical knowledge of web page design, JavaScript functions, arrays, form handling, DOM manipulation, and browser storage.

In conclusion, this project provides a useful introduction to web application development and can be improved further by adding additional features in the future.

References

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3. MDN Web Docs – JavaScript: https://developer.mozilla.org/en-US/docs/Web/JavaScript

4. MDN Web Docs – Web Storage API: https://developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API